

Honours Analysis 1 Tutorials (Sec. 003)

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Please note that this document has not been proofread and may contain typos. Feel free to see or email me if you have any questions.

ANNOUNCEMENT: I have moved my Thursday office hours to the 4:00pm-5:00pm slot. Please let me know if this causes any scheduling issues.

1 Eighth Tutorial

Let (x_n) be a sequence and suppose that x_n converges to $x \in \mathbb{R}$. It was seen in the lectures that any subsequence (x_{n_k}) of (x_n) must *also* converge to this point x . Let us also recall the following limit, which was discussed in class:

$$e := \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n. \quad (1.1)$$

We should also point out that this is our *definition* of e . Let us warm up by practicing with subsequences and limit-laws.

Example 1. We show that

$$\lim \left(1 - \frac{1}{n}\right)^n = \frac{1}{e}.$$

To see this, note that for $n \geq 2$:

$$\left(1 - \frac{1}{n}\right)^{-n} = \left(\frac{n-1}{n}\right)^{-n} = \left(\frac{n}{n-1}\right)^n.$$

Or, equivalently,

$$\left(1 - \frac{1}{n}\right)^{-n} = \left(1 + \frac{1}{n-1}\right)^n = \left(1 + \frac{1}{n-1}\right)^{n-1} \left(1 + \frac{1}{n-1}\right).$$

Therefore,

$$\begin{aligned}\lim \left(1 - \frac{1}{n}\right)^{-n} &= \lim \left(1 + \frac{1}{n-1}\right)^{n-1} \lim \left(1 + \frac{1}{n-1}\right) \\ &= \lim \left(1 + \frac{1}{n-1}\right)^{n-1} \\ &= e,\end{aligned}$$

where we have used (1.1) in this last step. Now, we write

$$\lim \left(1 - \frac{1}{n}\right)^n = \lim \frac{1}{\left(1 - \frac{1}{n}\right)^{-n}} = \frac{1}{\lim \left(1 - \frac{1}{n}\right)^{-n}} = \frac{1}{e}.$$

Example 2. Similarly, we will show that

$$\lim \left(1 + \frac{1}{2n}\right)^n = \sqrt{e}.$$

Here, we will instead make use of subsequences. For each $n \in \mathbb{N}$, we can write

$$\left(1 + \frac{1}{2n}\right)^n = \left[\left(1 + \frac{1}{2n}\right)^{2n}\right]^{\frac{1}{2}}. \quad (1.2)$$

Now,

$$\left(1 + \frac{1}{2n}\right)^{2n}$$

is a subsequence of $\left(1 + \frac{1}{n}\right)^n$, which we know to be convergent with limit e . This tells us that the sequence given above converges:

$$\lim \left(1 + \frac{1}{2n}\right)^{2n} = \lim \left(1 + \frac{1}{n}\right)^n = e.$$

Using this in (1.2), the limit laws tell us that

$$\lim \left(1 + \frac{1}{2n}\right)^n = \lim \left[\left(1 + \frac{1}{2n}\right)^{2n}\right]^{\frac{1}{2}} = \sqrt{e}.$$

Exercise. Find the flaw in the following “proof” that

$$\lim \left(1 + \frac{2}{n}\right)^n = e^2.$$

“*Proof*”. For every $n \in \mathbb{N}$ we can write

$$\left(1 + \frac{2}{n}\right)^n = \left[\left(1 + \frac{1}{n/2}\right)^{n/2}\right]^2.$$

Since we know that

$$\lim \left(1 + \frac{1}{n}\right)^n = e,$$

we get that

$$\lim \left(1 + \frac{1}{n/2}\right)^{n/2} = e.$$

Thus,

$$\lim \left(1 + \frac{2}{n}\right)^n = \lim \left[\left(1 + \frac{1}{n/2}\right)^{n/2}\right]^2 = e^2.$$

□

What exactly goes wrong with the proof above? Well, everything is correct up until the statement

$$\lim \left(1 + \frac{1}{n/2}\right)^{n/2} = e.$$

This does **not** follow from the fact that

$$\lim \left(1 + \frac{1}{n}\right)^n = e,$$

Indeed, although it seems to be at a first glance,

$$\left(1 + \frac{1}{n/2}\right)^{n/2}$$

is not a subsequence of

$$\left(1 + \frac{1}{n}\right)^n.$$

This is because $n/2$ is not be a natural number for every $n \in \mathbb{N}$.

1.1 Approximating square roots

Let $a > 0$ be given; we will construct a sequence in $\mathbb{R}$ converging to $\sqrt{a}$. In fact, as we shall soon see, this sequence will not require that we know the numerical value of $\sqrt{a}$. As a first step, pick **any** $x_1 > 0$ from $\mathbb{R}$. For $n \geq 1$, we define

$$x_{n+1} := \frac{1}{2} \left(x_n + \frac{a}{x_n} \right) > 0 \quad (1.3)$$

This gives rise to a recursive sequence (x_n) , with $x_n > 0$ for every $n \in \mathbb{N}$. As a first step, we will show that $x_n^2 \geq a$ for all $n \geq 2$. For any n , note that x_n is a root of the following equation:

$$x^2 - 2x_{n+1}x + a = 0.$$

In particular, the quadratic equation above has a real root. It follows from the quadratic formula that the discriminant

$$4x_{n+1}^2 - 4a$$

must be non-negative, i.e. $4x_{n+1}^2 - 4a \geq 0$. Thus, $x_{n+1}^2 \geq a$ whence $x_{n+1} \geq \sqrt{a}$.

As a second step, we show that the sequence (x_n) is decreasing for all $n \geq 2$. Namely, we show that $x_{n+1} \leq x_n$, whenever $n \geq 2$. To see this, we simply calculate

$$\begin{aligned} x_n - x_{n+1} &= x_n - \frac{1}{2} \left(x_n + \frac{a}{x_n} \right) = \frac{x_n}{2} - \frac{a}{2x_n} \\ &= \frac{1}{2} \left(\frac{x_n^2 - a}{x_n} \right) \geq 0. \end{aligned}$$

Here, we have used that $x_n^2 \geq a$ whenever $n \geq 2$. Thus, $x_{n+1} \geq x_n$ whenever $n \geq 2$. Combined, we have shown that (x_n) is an eventually decreasing sequence with $x_n \geq \sqrt{a}$ for $n \geq 2$. By the monotone convergence theorem, we know that (x_n) will converge to some $x \geq a$. In fact, using the limit laws with (1.3) shows that x will satisfy

$$x = \frac{1}{2} \left(x + \frac{a}{x} \right) \geq 0.$$

Rather,

$$x = \frac{a}{x} \geq 0.$$

Thus, we have $x^2 = a$ as was required. This recursive procedure is in general quite fruitful and simple. For this reason, we give one final example before moving on to more general results.

Example 3. Let $x_1 \geq 2$ be given and define for $n \geq 1$:

$$x_{n+1} := 1 + \sqrt{x_n - 1}. \quad (1.4)$$

We will show that the recursive sequence (x_n) converges and calculate its limit. First, let us show (by induction) that $x_n \geq 2$ for all $n \in \mathbb{N}$. The base case $n = 1$ is trivial. Assuming that $x_n \geq 2$, we then have

$$x_{n+1} = 1 + \sqrt{x_n - 1} \geq 1 + \sqrt{2 - 1} = 2.$$

Here, we have used that the function $f(x) := \sqrt{x}$ is monotone increasing on $[0, \infty)$.¹ We conclude that $x_n \geq 2$ for all $n \in \mathbb{N}$. In particular, $x_n - 1 \geq 1$ for all $n \in \mathbb{N}$. Thus, given $n \in \mathbb{N}$, we have

$$x_{n+1} = 1 + \sqrt{x_n - 1} \leq 1 + (x_n - 1) = x_n.$$

In this last step, we have used that $\sqrt{t} \leq t$ for all $t \geq 1$.² Hence, (x_n) is a monotone decreasing sequence that is bounded below by 2. It therefore converges to some point $x \geq 2$. Using the limit laws and (1.4), this point x must satisfy

$$x = 1 + \sqrt{x - 1}.$$

It follows that $(x - 1) = \sqrt{x - 1}$, where $x - 1 \geq 1$. Hence, $\sqrt{x - 1} = 1$ whence $x = 2$. We conclude that $\lim x_n = 2$.

¹To see why this is true, suppose that $0 \leq x < y$. Then, we calculate

$$\sqrt{y} - \sqrt{x} = \frac{(\sqrt{y} - \sqrt{x})(\sqrt{y} + \sqrt{x})}{\sqrt{y} + \sqrt{x}} = \frac{y - x}{\sqrt{x} + \sqrt{y}} \geq 0.$$

Thus, $\sqrt{x} \leq \sqrt{y}$ whenever $0 \leq x < y$. Since this also holds for $0 \leq x = y$, we see that $f(x) = \sqrt{x}$ is monotone increasing on $[0, \infty)$.

²To justify this inequality, let $t \geq 1$ and note that

$$t - \sqrt{t} = \sqrt{t}(\sqrt{t} - 1) \geq \sqrt{t} - 1 \geq 0.$$

1.2 The Bolzano-Weierstrass Theorem

In this section, we prove one of the most fundamental and useful theorems in real analysis. Although the result heavily hinges upon our cumulative efforts thus far, the theorem will easily follow from one simple observation.

Definition 1. Let (x_n) be a sequence of real numbers. The N^{th} term x_N is called a *peak* of the sequence (x_n) if, for every $n \geq N$, there holds $x_n \leq x_N$.

In light of this definition, we have a relatively simple proof of our key lemma.

Lemma 1. *Any sequence of real numbers has a monotone subsequence.*

Proof. Let (x_n) be a sequence of real numbers. We distinguish the only two possible cases.

1. Suppose (x_n) has *infinitely many* peaks. Then, we can find a subsequence (x_{n_k}) of (x_n) consisting entirely of peaks. For every index k , x_{n_k} is a peak of the sequence (x_n) . In particular, x_{n_1} is a peak of the sequence (x_n) . Since $n_2 > n_1$, we must then have $x_{n_1} \geq x_{n_2}$. Similarly, since x_{n_2} is a peak and $n_3 > n_2$, there holds

$$x_{n_2} \geq x_{n_3}$$

and so forth. In general, we have

$$x_{n_1} \geq x_{n_2} \geq \cdots \geq x_{n_k} \geq \cdots$$

That is, (x_{n_k}) is a monotone subsequence of (x_n) .

2. Otherwise, (x_n) has only finitely many peaks. Thus, we can find $N \in \mathbb{N}$ so large that x_n is not a peak of the sequence (x_n) for all $n \geq N$. Define $x_{n_1} := x_N$. Since x_{n_1} is not a peak of the sequence (x_n) , there is some x_j in our sequence, with $j \geq n_1$, such that $x_j > x_{n_1}$. In particular, $j \neq n_1$. Put then

$$x_{n_2} := x_j$$

so that $x_{n_1} < x_{n_2}$ and $N \leq n_1 < n_2$. Again, x_{n_2} is not a peak of the sequence (x_n) . By a similar argument, we can find $j > n_2$ such that $x_j > x_{n_2}$. Define then $x_{n_3} := x_j$ so that

$$x_{n_1} < x_{n_2} < x_{n_3}.$$

Continuing in this way, we construct a monotone increasing subsequence (x_{n_k}) of (x_n) .

In either case, we have found a monotone subsequence. $\square$

Armed with this lemma, we can easily establish the following powerhouse of a theorem.

Theorem 1.1 (Bolzano-Weierstrass). *Any bounded sequence in $\mathbb{R}$ has a convergent subsequence.*

Remark 1.1. The theorem above continues to hold in $\mathbb{R}^m$ and $\mathbb{C}^n$ for all $m, n \in \mathbb{N}$. Such theorems are known as *compactness* theorems and are extremely powerful.

Proof of Theorem. By our lemma, any sequence has a monotone subsequence. If (x_n) is bounded, then any monotone subsequence of (x_n) is also bounded. By the monotone convergence theorem, such a subsequence converges. $\square$

1.3 More about the Supremum

Proposition 2. *Let $E \subseteq \mathbb{R}$ be non-empty and bounded from above. Fix a point $y \in \mathbb{R}$; the following statements are equivalent:*

- (1) *y is the limit of two sequences (x_n) and (y_n) such that, for every $n \in \mathbb{N}$, $x_n \in E$ and y_n is an upper bound of E .*
- (2) *$y = \sup E$.*

Proof. Suppose that (1) holds. For every $n \in \mathbb{N}$, y_n is an upper bound for E and $x_n \in E$. Therefore, we must have

$$x_n \leq \sup E \leq y_n$$

for each $n \in \mathbb{N}$. Letting $n \rightarrow \infty$, the Squeeze Theorem implies that $y \leq \sup E \leq y$.

Conversely, let $y := \sup E$. For every $n \in \mathbb{N}$, the quantity $y - \frac{1}{n}$ is no longer an upper bound for E (y is the least upper bound). Thus, there exists $x_n \in E$ such that

$$y - \frac{1}{n} < x_n \leq y.$$

This gives us a sequence (x_n) in E with $\lim x_n = y$. Now, take (y_n) to be the constant sequence $y_n := y$. $\square$

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